

MSAS – Assignment #2: Modeling

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Exercise 1

To perform a preliminary analysis and evaluate a proposed satellite configuration, it is requested to model an active thermal control system and its associated mechanisms to assess the system's capability to maintain the temperatures within a certain range $[T_{min}; T_{max}]$. The satellite body is a rectangular cuboid (1.5 [m] height, 0.5 [m] side), with fixed solar panels (0.5 [m] side, 0.95 [m] length) located on the $+Y$ -axis. The satellite flies with a fixed inertial attitude, with the satellite's Y -axis always directed towards the Sun. The satellite has two extendable radiators, which are located on the $\pm X$ -axis faces. The radiators are connected to the satellite body with electrically controlled hinges that can rotate along the Z -axis direction. The satellite with its radiators in open and closed configurations is represented in Figure 1.

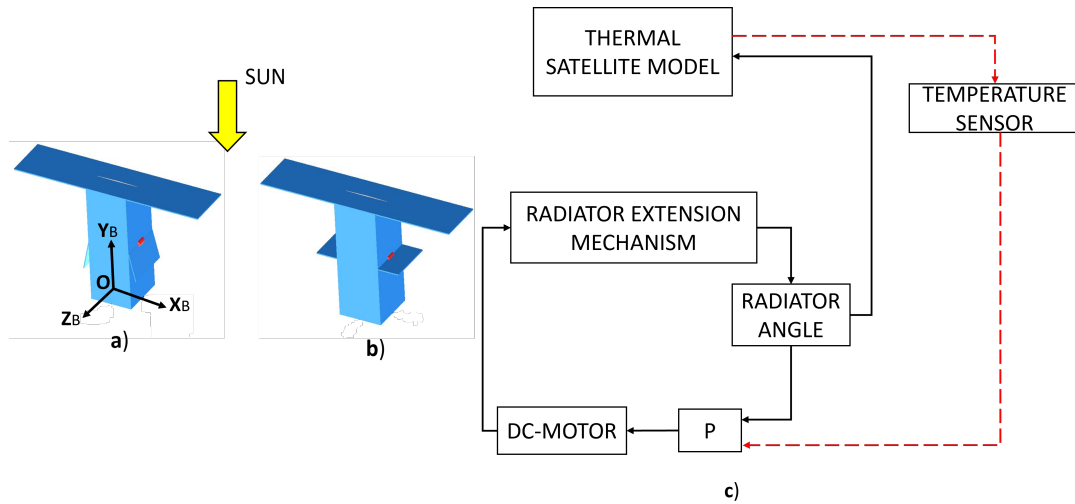


Figure 1: Satellite configuration: a) Closed radiator; b) Open radiator; c) Thermal control logic overview

The extension mechanism is based on a DC motor which gives the needed torque to move the radiator to the desired position. In Figure 2 the physical model of the mechanism is shown. The angle θ represents the rotation of the radiator with respect to the Z -axis and is measured counter-clockwise. For a rotation angle $\theta = 0$ deg, the radiators are in fully open configuration, for $\theta = -0.4\pi$ the radiators are in stowed (closed) configuration. The main parameters of the electro-mechanical model are reported in Table 1.

Parameter	Symbol	Value	Units
Resistance	R	0.1	Ω
Inductance	L	0.001	H
Motor constant	k_m	0.3	Nm/A
Radiator mass	m_r	0.2	kg
Radiator length	L_r	0.5	m

Table 1: Characteristics of the thermal control mechanism.

The radiator has a lower emissivity side, facing deep space in the closed configuration, and a higher emissivity side which is hidden when the radiator is in the stowed position. It is assumed

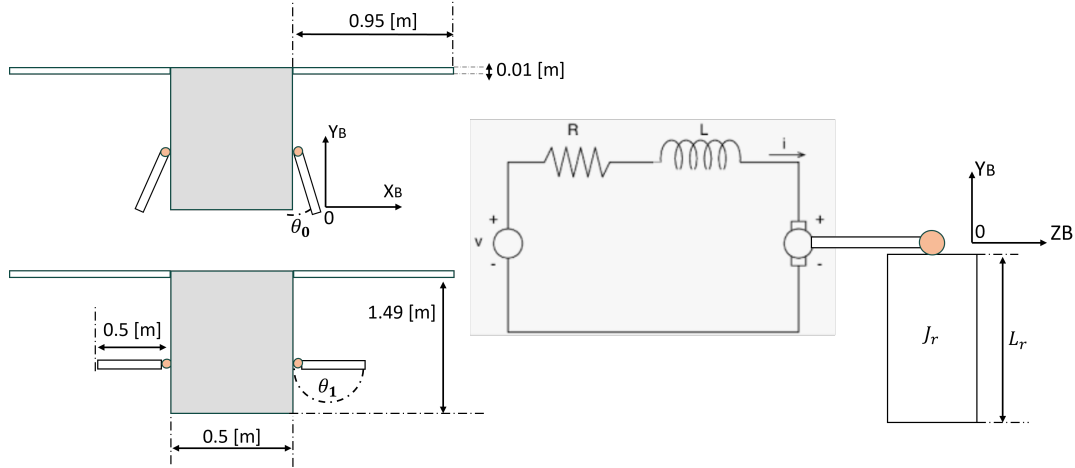


Figure 2: Schematics of the radiator extension mechanism

that radiative heat exchange among the satellite surfaces (re-radiation) is negligible and that no other thermal radiation sources (e.g., planets) are present. As a result, the thermal control of the satellite is regulated mainly by the radiator angle θ , as shown in Figure 1 c). By opening and closing the radiator, the effective area of the high- and low-emissivity sides exposed to deep space increases and decreases respectively. This results in a linear variation of the emissivity ϵ as a function of θ that can be expressed as in Eq. (1):

$$\epsilon(\theta) = \epsilon_{min} + \left[\frac{\epsilon_{max} - \epsilon_{min}}{0.4 \cdot \pi} \right] (\theta(t) + 0.4 \cdot \pi) \quad (1)$$

When the radiator is in a closed configuration with $\theta(t) = -0.4\pi$ [rad] the emissivity is at its minimum $\epsilon_{min} = 0.01$. In fully open configuration instead, when $\theta(t) = 0$ [rad], the radiator emissivity reaches its maximum value $\epsilon_{max} = 0.98$. Using a lumped nodes thermal modeling, the satellite can be divided into 5 components (the two solar panels, the main body, and the two radiators). The radiators and solar panels are connected to the main body node by a conductive path, while all nodes are thermally radiating towards deep space (see Figure 3).

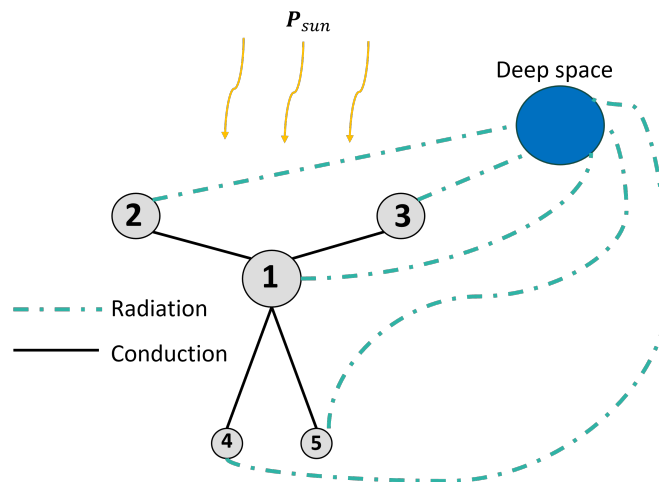


Figure 3: The thermal lumped nodes model

The solar radiation from the Sun is acting only on the two solar panel nodes and on the top face of the main-body node (the radiators are always in shadow). The temperature of deep space is constant and equal to $T_{ds} = 3$ [K]. The characteristics of the thermal network and its components are given in Table 2.

Parameter	Symbol	Value	Unit
Sun Power	P_{Sun}	1350	W/m ²
Heat capacity	C_1	1.5×10^5	J/K
Heat capacity	C_2, C_3	1187.5	J/K
Heat capacity	C_4, C_5	30	J/K
Thermal conductance	$G_{12}, G_{13}, G_{14}, G_{15}$	10	W/K
Absorptivity	α_1	0.6	-
Absorptivity	α_2, α_3	0.78	-
Emissivity	ϵ_1	0.45	-
Emissivity	ϵ_2, ϵ_3	0.75	-

Table 2: Thermal parameters of the system.**Part 1: causal modeling (9 points)**

Considering the following main constitutive equations of the system:

$$C_i \frac{dT_i}{dt} = \sum_i^n (Q_{in}^i - Q_{out}^i) \quad (2)$$

$$V_{in} = k_p \cdot (T_1 - T_1^{ref}) \quad (3)$$

$$J_r \ddot{\theta} = \tau_{in} \quad (4)$$

where Eq. (2) is the conservation of energy at a thermal node; Eq. (3) is a proportional control law in which V_{in} is the input voltage applied on the DC-motor, and T_1^{ref} is the reference temperature of the main-body; and Eq. (4) is the simplified rotational radiator dynamics where the input torque is linked to current flowing into the motor through the relation: $\tau_{in} = k_m \cdot i$. Assuming that: the upper limit $\hat{T}_{max}$ and lower limit temperature $\hat{T}_{min}$ allowed for **node 1** are 300 [K] and 290 [K] respectively, the motor control is capable of maintaining the angle θ within the physical limits, all the thermal nodes start with a temperature of 298.15 [K], and that the radiator is closed at the beginning of the simulation ($\theta(0) = -0.4\pi$):

1. Formulate the full system of non-linear ODEs making explicit the state variables of the whole simulation.
2. Considering a simulation time of at least 50 [h], define the value for the proportional gain k_p such that: the temperature is kept around the target $T_1^{ref} = 294.15$ [K], and within $t = 10$ [h] the maximum temperature oscillations are less than 0.1% of T_1^{ref} . Plot the resulting temperature evolution over time on all thermal nodes.
3. Discuss the results and the ode used in Matlab for the integration.

Part 2: acausal modeling (6 points)

Using the Modelica standard library, reproduce in Dymola the physical model of the thermal-electro-mechanical system described above. You can build your **own model block** in Modelica to implement specific signals such as $\epsilon(\theta)$ or the control strategy. Please note that when you create a new model block (see in Table 3) you can insert input/output variables and your equations (see as an example the **Text View** of the block: *Modelica/Blocks/Math/Add*). Then, simulate it in OpenModelica comparing the results with the ones obtained using the causal model.

(15 points)

<i>Physical element</i>	<i>Modelica library</i>
Rigid-body	Modelica/Mechanics/MultiBody/Parts
Revolute joint	Modelica/Mechanics/MultiBody/Parts
Fixed-translational element	Modelica/Mechanics/MultiBody/Parts
DC-motor	Modelica/Electrical/Machines/BasicMachines/DCMachines
Thermal node	Modelica/Thermal/HeatTransfer/Components
Thermal resistances	Modelica/Thermal/HeatTransfer/Components
Logical switch	Modelica/Blocks/Logical/Switch
Hysteresis	Modelica/Blocks/Logical/Switch
Temperature sensor	Modelica/Thermal/HeatTransfer/Sensors
Your model block	File/New/Block

Table 3: Modelica libraries.**Part 1: Causal Modeling****Point 1:**

The system is modeled as three interacting subsystems: Mechanical, Electrical, and Thermal, each described by its respective ODEs.

1. Mechanical Subsystem In the mechanical subsystem are considered the motions followed by the radiators, therefore angular velocity, ω , and angular displacement, θ , undergone are written in Eq. 5 and 6.

$$\dot{\theta} = \omega \quad (5)$$

$$\dot{\omega} = \ddot{\theta} = \frac{k_m \cdot I}{J_r} \quad (6)$$

The moment of inertia, J_r , is calculated using Eq. 7, assuming a thin rectangular plate geometry rotating about an edge ($L = 0.5$).

$$J_r = \frac{1}{3} m_r L_r \quad (7)$$

2. Electrical Subsystem

The electrical system is considered to have the state variable I , representing the current passing through the circuit. Its corresponding equation is given in Eq. 8, in which L stands for inductance and V_{in} is the input voltage given by Eq. 3.

$$\dot{I} = \frac{V_{in} - I \cdot R - k_m \cdot \theta}{L} \quad (8)$$

3. Thermal Subsystem

The thermal subsystem includes all the thermal nodes, bodies, and the variation of their temperatures. The thermal subsystem is characterized by the following equations, where $\sigma = 5.67 \cdot 10^{-8} \frac{W}{m^2 K^4}$ is the Stefan-Boltzmann constant.

$$\begin{aligned} \dot{T}_1 = \frac{1}{C_1} (P_{Sun} \cdot \alpha_1 \cdot A_{abs1} + G_{12} \cdot (T_2 - T_1) + G_{13} \cdot (T_3 - T_1) + G_{14} \cdot (T_4 - T_1) + \\ + G_{15} \cdot (T_5 - T_1) - \epsilon_1 \cdot \sigma \cdot A_{emiss1} \cdot (T_1^4 - T_{ds}^4)) \end{aligned} \quad (9)$$

$$\dot{T}_2 = \frac{1}{C_2} (P_{Sun} \cdot \alpha_2 \cdot A_{abs2} + G_{12} \cdot (T_1 - T_2) - \epsilon_2 \cdot \sigma \cdot A_{emiss2} \cdot (T_2^4 - T_{ds}^4)) \quad (10)$$

	Absorptivity[m ²]	Emissivity[m ²]
Main Body	0.25	3.25
Panels	0.475	0.475
Radiators	-	0.25

Table 4: Absorptivity and Emissivity Areas

$$\dot{T}_3 = \frac{1}{C_3} (P_{Sun} \cdot \alpha_3 \cdot A_{abs3} + G_{13} \cdot (T_1 - T_3) - \epsilon_3 \cdot \sigma \cdot A_{emiss3} \cdot (T_3^4 - T_{ds}^4)) \quad (11)$$

$$\dot{T}_4 = \frac{1}{C_4} (G_{14} \cdot (T_1 - T_4) - \epsilon \cdot \sigma \cdot A_{emiss4} \cdot (T_4^4 - T_{ds}^4)) \quad (12)$$

$$\dot{T}_5 = \frac{1}{C_5} (G_{15} \cdot (T_1 - T_5) - \epsilon \cdot \sigma \cdot A_{emiss5} \cdot (T_5^4 - T_{ds}^4)) \quad (13)$$

In the above equations, A_{abs} represents the area of each thermal node that absorbs heat, while A_{emiss} is the emissivity area for each of the thermal nodes, considering that the nodes' numbering was kept similar to that in Fig. 3. The areas are reported in Tab. 4. It was considered that the Main Body absorbs heat only from the upper panel, and it emits heat through the rest of the body, while for the panels they absorb the Sun heat similarly from the upper body, and emits it only through lower body. Moreover, the problem appears to be symmetrical through the panels and the radiators. The symmetrical design and solar heat exposure of the panels result in identical temperature evolution during the simulation leads to an identical value and evolution for the two bodies. Similarly, for the radiators, T_4 and T_5 have identical variations. The state variables of the system are the temperatures T_1, T_2, T_3, T_4, T_5 , the current I and the rotation and angular velocity, θ and ω , for which the system is solved over the time interval using an integrator.

Each emissivity term in the equations is negative as the nodes change heat with deep space. The 1st node has conduction terms in opposite direction with respect to the other bodies because, while it absorbs heat, the other bodies transfer heat to node 1.

Point 2:

The gain k_p was optimized through a trial-and-error approach to meet the temperature requirements for T_1 . A large k_p decreases the oscillation wavelength, resulting in a very oscillating behavior, imposing a large mechanical stress on the system. Reducing its value solves the problem, and thus it was considered to have $k_p = 10^{-4}$.

The initial conditions for the system, besides those already given in the text were chosen to be $I = 0$ and $\omega = 0$, such that the radiators are static at the beginning of the simulation, and start to move when the simulation starts.

In order to make the temperature of node 1, to converge towards the reference value of 294.15 and to be in range of around $\pm 0.1\%$ after 10 h of simulation, it was considered that the radiators should remain fully opened until the temperature of node 1 is smaller than T_{ref} . Thus, if these two conditions were met, $\theta \geq 0$ and $T_1 > T_{ref}$ the angular velocity($\dot{\theta}$) of the radiator would be imposed to be 0 and its position fixed at maximum value. This can be seen in Fig. 5.

A similar condition was implemented in the case when θ reaches its minimum value(-0.4π) and $T_1 < T_{ref}$. From the evolution of $T_1(t)$ is observed that the temperature requirements of the system are satisfied, T_1 is within the desired bounds at $t = 10$ h.

Similar, to Fig. 5, Fig. 4 illustrates an oscillatory behavior of the system. This is explained as each time the temperature of node 1 reaches T_{ref} , ω changes direction, so that the desired convergence is obtained. The temperature converges until the end of the time interval, due to

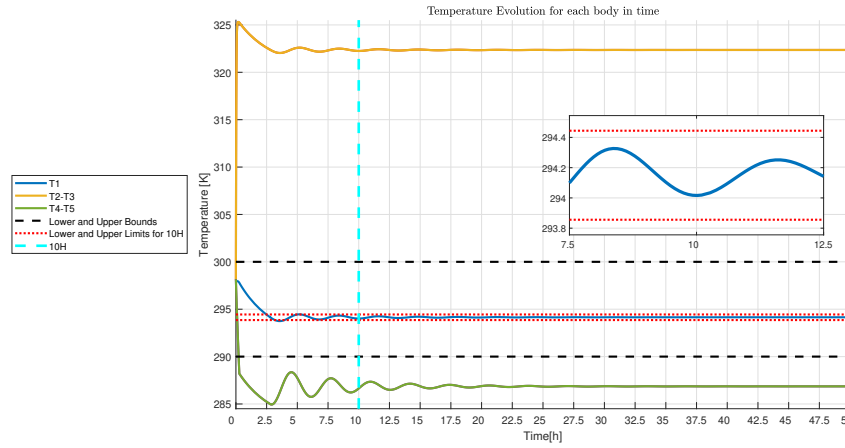


Figure 4: Temperature evolution of each thermal node

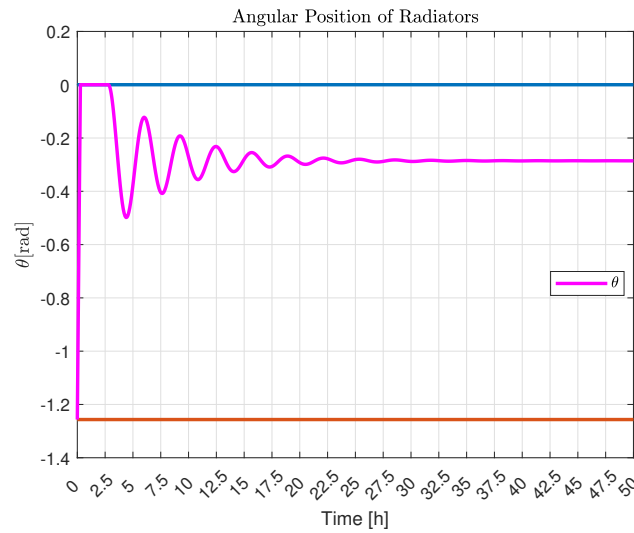


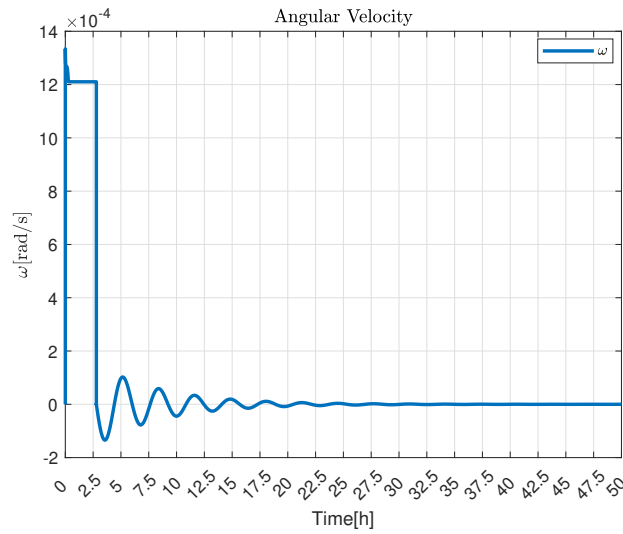
Figure 5: Radiators' angular position

very small changes of θ , the system almost reaching an equilibrium state. This state is reached at around $\theta = -0.0909\pi$ and $T_1 = 294.1501$, which can be observed in Fig. 6. The radiators reach their maximum position in approximately 16 minutes without inducing significant mechanical stress, and they are kept fixed in this position for 2:30 hours, or until the temperature drops below the reference one.

Point 3:

The equations given in **Point 1** are placed in *ode15s*. This integrator was chosen, as the system of equations was evaluated to be stiff. This is understandable as the characteristic time-scale of the electrical subsystem is smaller than that of the mechanical subsystem, while the thermal one expected to have the smallest characteristic time-scale. Analysis of computational times confirmed the system's stiffness, necessitating the use of a stiff solver, therefore it was used *ode15s* to obtain computational efficiency. The times are reported in Table 5.

Regarding the integration, the tolerances were set to be 10^{-6} , as they offer a good approximation, and the code continues to run smoothly. This was chosen based on the order of magnitudes of angular velocity. As this state variable is of the order of magnitude of 10^{-6} . Moreover, a small tolerance is desired to stop radiators close to their limits. As the control is

**Figure 6:** Radiators' angular velocity evolution

ODE	Time [s]
<i>ode15s</i>	0.85
<i>ode78</i>	565.7
<i>ode113</i>	569.2
<i>ode45</i>	522.79

Table 5: Times for each integrator

imposed only once θ passes the maximum/minimum limit, a smaller tolerance will enable the system to stop the radiators as close to the limit as desired, while maintaining an acceptable computational efficiency.

Part 2: Acausal Modeling

For the acausal modeling the above thermal-electro-mechanical system was implemented in Dymola as shown in Fig. 7.

The scheme is divided in 5 regions, each represented by a different color. The yellow blocks represent the solar panels, the blue blocks represent the radiator, the green block represents the control subsystem while the two blocks in beige reflect the main body with both thermal and electric subsystem.

For the Solar Panels, the heat absorbed by them was simulated through a *FixedHeatFlowSource*, whose value was selected to be $P = \alpha \cdot A_{abs} \cdot P_{Sun}$. For the main body, the power absorbed was similarly implemented, within the central *FixedHeatFlowSource* to be 202.5. 5 temperature sources were included for clarity of the scheme, as 1 block would suffice for the simulation, connected accordingly to the desired blocks. Each body is represented through a *HeatCapacitor*, while their thermal interaction is done through a *ThermalConductor*. Two different approaches were taken for the implementation of emissivity ϵ into the system. For the bodies with fixed emissivity, such as the main body and the solar panels, this was included into the *BodyRadiation* block. For the 2 radiators the emissivity is introduced into the system through a block called *Emissivity*. This takes the heat flows multiply it with ϵ and as output has the final heat flow. An angle sensor is placed after the *Inertia* block to provide the angle into the *AngletoEmissivity* block, which transforms the angle into emissivity with respect to the Eq. 1.

The system is similar in all other aspects with the equations from the causal modeling with

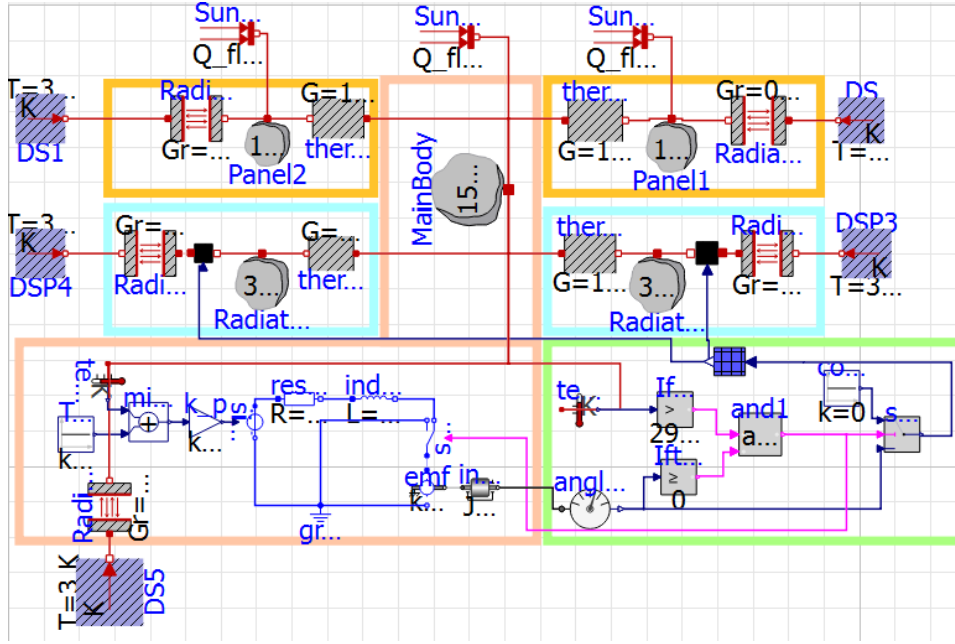
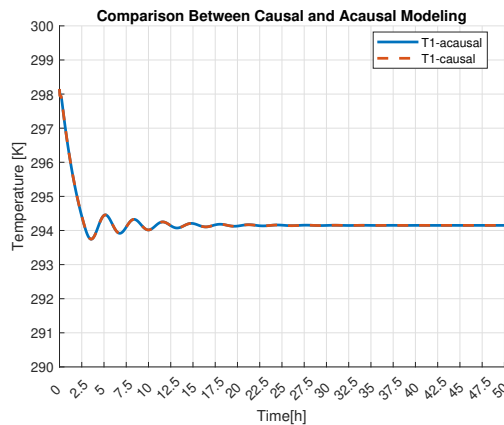


Figure 7: Dymola Scheme of the system

one exception. In the causal modeling a switch was implemented in the electric circuit, in order to set the current passing through the DC motor to zero. Compared to the causal case, now the condition of the radiator angle cannot be set to stop after the DC motor, as the angle is set by the current passing through the circuit. Thus, following Eq. 6, the current passing through the system is set to zero, through the gate connected to the ground, so that $\dot{\theta} = 0$. This is used for the control subsystem. In this also θ is set to zero, if $T_1 \leq 294.15$ and $\theta \geq 0$.

The results are similar to the ones obtained from the causal modeling as shown in Fig. 8a and 8b. From Fig. 8a it can be observed that the temperature of the main body(1) converges to $T = 294.15$ and is equal to the one obtained from the causal model.

Similarly, the evolution of θ angle is equal to the one obtained from the MATLAB as shown in Fig. 8b.



(a) Difference between causal and acausal temperature evolution

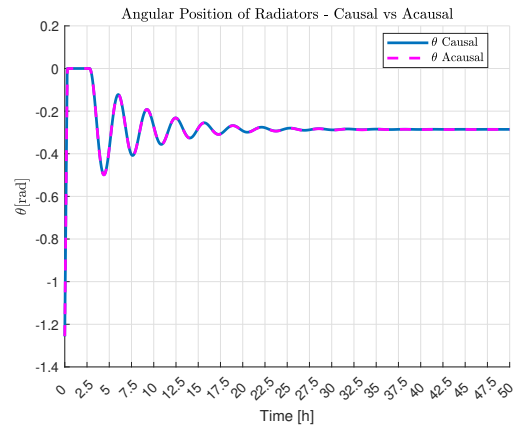

 (b) Difference between causal and acausal θ evolution

Figure 8: Differences Between Causal and Acausal Model

Exercise 2

A simplified conceptual map of the Attitude Control System (ACS) of a satellite is depicted in Figure 9. For the accuracy of some measurements that the spacecraft has to obtain, the probe has to fly in an orbit very close to the Earth's surface. The spacecraft is therefore subjected to a high atmospheric drag. The ACS has to compensate for the deceleration caused by this drag continuously. In general, ACSs are made of 3 sensors in orthogonal directions and 3 thrusters to detect and compensate for linear and angular accelerations on the probe in each direction, but for the sake of this project, only compensation in the tangential direction will be considered.

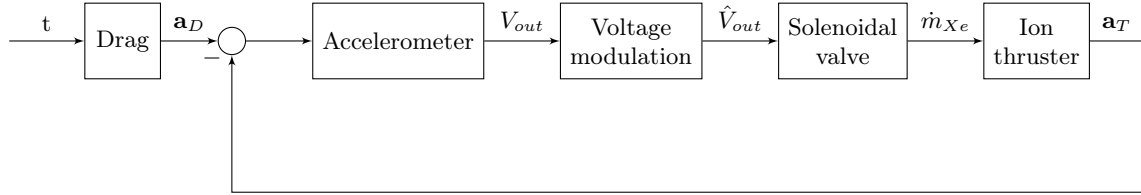


Figure 9: Block diagram of the ACS.

The ACS is modeled with several modules: an **accelerometer** measuring the accelerations acting on the spacecraft, a **voltage modulation block** controlling the output voltage, the **control valve** modifying the aperture of the thruster valve, and the **ion thruster**.

The accelerometer, kept aligned with the direction of velocity, presents inside a seismic mass that is subject to an external acceleration $\mathbf{a}_D$ and $\mathbf{a}_T$ given only by the drag D and the thrust T , respectively. The mass is in between the stators of a condenser and, moving for the acceleration given by these two forces changes the output voltage V_{out} of the circuit of the accelerometer. The dynamics of the seismic mass itself can be modeled as a mass-spring-damper system as follows

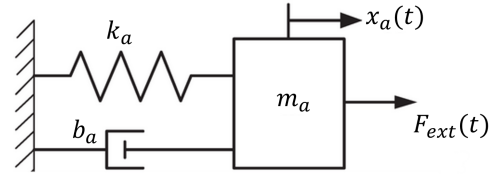


Figure 10: Model of the accelerometer.

$$\dot{x}_a = v_a \quad (14)$$

$$F_{ext}(t) = \frac{T - D}{M_{SC}} m_a = m_a \dot{v}_a + b_a v_a + k_a x_a \quad (15)$$

where M_{SC} is the mass of the spacecraft, m_a is the seismic mass value, and b_a and k_a are respectively the damper and the spring terms present in the accelerometer. For the causal modeling, consider the V_{out} as directly proportional to the velocity of the seismic mass, as $V_{out} = K_{acc} v_a$ [1].

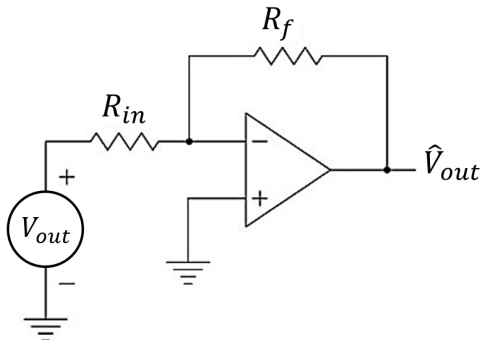


Figure 11: Inverting operational amplifier.

The output of the accelerometer V_{out} needs to be modulated into $\hat{V}_{out}$ to adjust the control for the solenoidal valve. This modulation is performed through an operational amplifier, displayed in Figure 11. The operational amplifier in inverting configuration modifies V_{out} into $\hat{V}_{out}$ accordingly to:

$$\hat{V}_{out} = -\frac{R_f}{R_{in}} V_{out} \quad (16)$$

where R_f and R_{in} are the two resistances modifying the input voltage.

The voltage $\hat{V}_{out}$ is itself the input of the solenoidal valve, shown in Figure 12, and creates a current I commanding the flow control valve. The current passes through a circuit modeled with only a solenoid with a variable inductance $L(x_v)$. As the current variably flows, a magnetic field is generated. A resulting force f_v acts on the spool, which as a result moves.

The armature-spool arrangement is then modeled again as a lumped parameter spring-mass-damper system

$$\dot{x}_v = v_v \quad (17)$$

$$m_v \dot{v}_v = -k_v x_v - b_v v_v + \frac{1}{2} I^2 \frac{dL}{dx_v} \quad (18)$$

$$\dot{I} = \frac{1}{L(x_v)} \hat{V}_{out} \quad (19)$$

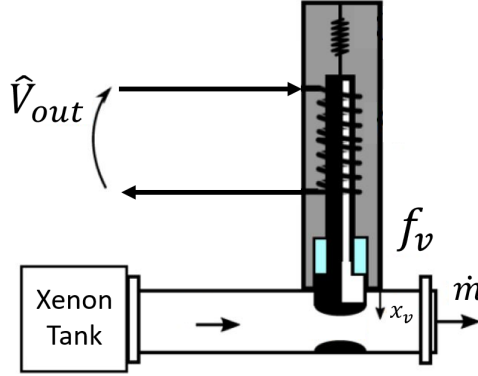


Figure 12: Model of the solenoidal valve.

where x_v and v_v represent the position and velocity of the spool, and b_v and k_v are respectively the damper and the spring terms, and m_v is the mass of the spool. The dynamics of the current I is generated by $\hat{V}_{out}$ through the variable inductance $L(x_v) = \frac{1}{\alpha + \beta x_v}$, depending on the position of the valve itself.

As the spool moves, the area $A_v(x_v)$ of the duct through which a Xenon flow is passing changes. The resulting mass flow rate $\dot{m}_{Xe}$ of Xenon is ejected into the ion thruster, shown in Figure 13. In particular, the area of the valve can be modeled as

$$A_v(x_v) = A_0 + \ell(x_{v,\max} - x_v) \quad (20)$$

In particular, the orifice area $A_v(x_v)$ has been modeled as linearly varying with x_v . The tool controlling the area is a flap of width ℓ and maximum extension $x_{v,\max} = \ell$, for simplicity.

The value A_0 is the minimum area of the orifice, such that:

$$0 \leq x_v \leq x_{v,\max}$$

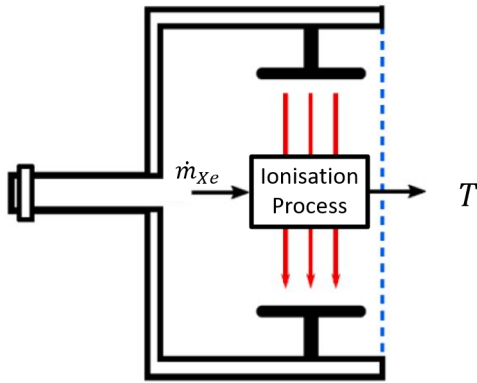


Figure 13: Model of the ion thruster.

electric field, producing a thrust on the spacecraft as the final effect. The flux of Xenon $\dot{m}_{Xe}$ can be expressed as follows

$$\dot{m}_{Xe} = A_v(x_v) \sqrt{k \rho_{Xe} p_T \left(\frac{2}{k+1} \right)^{\frac{k+1}{k-1}}} \quad (22)$$

where $\rho_{Xe} = \frac{p_T}{\bar{R} T_T}$, k and $\bar{R}$ being respectively Xenon specific heat ratio and gas constant. In the tank, the gas is at a total pressure of p_T and temperature of T_T . The Xenon flow enters the

thruster and is here ionized into ions with mass m_i and charge q . After the acceleration imposed by the acceleration grid with a voltage ΔV , the ions exit from the nozzle with a velocity v_{exit} , producing the thrust T , as follows

$$T = \dot{m}_{Xe} v_{exit} = \dot{m}_{Xe} \sqrt{\frac{2q\Delta V}{m_i}} \quad (23)$$

Table 6 reports the values for the parameters to simulate the ACS system, with symbol and unit of measure. The drag $D(t)$ can be modeled as a function of time only, as it follows

$$D(t) = 2.2 - \cos(\omega_s t) + 1.2 \sin(\omega_o t) \cos(\omega_o t) \quad (24)$$

where t enters in seconds, and $D(t)$ is in mN.

Component	Parameter	Symbol	Value	Unit
Accelerometer	Spacecraft mass	M_{SC}	300	kg
	Seismic mass	m_a	0.32	kg
	Accelerometer damper	b_a	$[1.5 \cdot 10^3 - 2 \cdot 10^4]$	Ns/m
	Accelerometer spring	k_a	$[5 \cdot 10^{-5} - 3 \cdot 10^{-3}]$	N/m
	Acc. proportional coefficient	K_{acc}	1	Vs/m
Amplifier	Inverting resistance	R_{in}	$[0.1 - 10]$	Ω
	Feedback resistance	R_f	$[1 \cdot 10^4 - 8 \cdot 10^4]$	Ω
Solenoidal Valve	Spool mass	m_v	0.1	kg
	Valve spring	k_v	$1 \cdot 10^3$	N/m
	Valve damper	b_v	$1 \cdot 10^3$	Ns/m
	Solenoid constant	α	$2.1 \cdot 10^{-2}$	1/H
	Solenoid gain	β	-60	1/Hm
	Minimum area	A_0	$4.7 \cdot 10^{-12}$	m ²
	Maximum extension	$x_{v,max}$	$1 \cdot 10^{-5}$	m
Thruster	Heat ratio	k	1.66	-
	Tank pressure	p_T	$2 \cdot 10^5$	Pa
	Tank temperature	T_T	240	K
	Gas constant	$\bar{R}$	63.32754	J/kg K
	Charge	q	$1.6 \cdot 10^{-19}$	C
	Voltage	ΔV	2000	V
	Ion mass	m_i	$2.188 \cdot 10^{-25}$	kg
Drag	Secular pulsation	ω_s	$1.658226 \cdot 10^{-6}$	rad/s
	Orbital pulsation	ω_o	$1.160758 \cdot 10^{-3}$	rad/s

Table 6: Parameters for the simulation of the ACS.

Part 1: causal modeling (9 points)

Reproduce in Matlab the physical model of the ACS, in particular

1. Formulate the full system of non-linear ODEs making explicit the state variables of the whole simulation. Tune the values of the parameters not explicitly stated in Table 6 to have the compensating thrust T the most similar to the disturbing drag D .
2. Choose the appropriate initial conditions and simulate for three orbital periods T_o , where $T_o = 2\pi/\omega_o$.
3. Discuss the selection of the ODE integration scheme.

Part 2: acausal modeling (6 points)

Reproduce in Simscape the physical model of the ACS. Choose the appropriate initial conditions and parameters, simulate for three orbital periods T_o , and compare the results with those obtained in the prior point.

Note: when modeling the solenoid valve in Simscape, please consider the nominal values shown in Table 7.

Parameter	Value	Unit
Pull-in forces $[F_1 - F_2]$	$[9000 - 12]$	N
Stroke $[x_1 - x_2]$	$[0 - 0.1]$	mm
Maximum Stroke	0.1	mm
Rated voltage	0.6	mV
Rated current	0.1	A
Contact stiffness	0.12×10^6	N/m
Contact damping	10^4	Ns/m

Table 7: Solenoid parameters in Simscape.

(15 points)

Part 1: Causal Modeling

Point 1: The system can be reduced to 5 ODEs and 5 state variables. The state variables are the state variables of the accelerometer, x_a , v_a and the state variables of the solenoid x_v , v_v and I . The other parameters are dependent on the evolution of these variables and evolve along these.

Therefore, the system of non-linear ODEs becomes:

$$\dot{x}_a = v_a \quad (25)$$

$$\dot{v}_a = \frac{T - D}{M_{SC}} - \frac{b_a \cdot v_a + k_a \cdot x_a}{m_a} \quad (26)$$

$$\dot{x}_v = v_v \quad (27)$$

$$\dot{v}_v = \frac{-k_v \cdot x_v - b_v \cdot v_v - \frac{1}{2} \cdot I^2 \cdot \frac{\beta}{(\alpha + \beta \cdot x_v)^2}}{m_v} \quad (28)$$

$$\dot{I} = -\frac{R_f}{R_{in}} \cdot k_{acc} \cdot v_a \cdot (\alpha + \beta \cdot x_v) \quad (29)$$

Regarding the initial conditions for the system, there were considered more small values through trial-and-error method and it was observed that taking initial values as 10^{-8} would lead to satisfactory results. More initial conditions were observed to lead to similar results, but in the end it was chosen the value of 10^{-8} , as it provides an order of magnitude at the middle of the orders of magnitude of the results. This was chosen as it is in the vicinity of all the state variables with very small orders of magnitude difference.

For the variables not explicitly stated in the text, the values in Table 8 were chosen. These were again chosen based on trial-and-error, and were the ones that lead to the most satisfactory results, leading to the closest values of T w.r.t. D , as requested. However, it must be noted that in the case of the resistances, R_f and R_{in} , other values would also lead to the same result, as long as the value of their fraction is kept constant. For example, also the values 0.2 and $2 \cdot 10^4$ would have the same result. However, R_{in} has a bigger impact on the system evolution than R_f , as for different values of the last the system's evolution presented small changes, that could be neglected, while changes along the interval in R_{in} would result in larger changes in the final outcome. This is expected, as for R_f the values along its interval remain at the

same order of magnitude while for R_{in} there are 3 different orders of magnitude encountered. The goal is to maximize $\hat{V}_{out}/V_{out}$, which is done by choosing the values reported in Table 8. Regarding the accelerometer parameters, the system is constrained to be overdamped, ($\xi_a > 1$), but the optimal values result in a smaller damping ratio for faster convergence, thus the values chosen for the accelerometer are reported in Table 8. It must be noted that these do not represent all the values of the variables k_a , b_a , R_{in} and R_f that would result in satisfactory solutions, however, these were chosen as they were the first found and the easiest to compute.

Parameter	Value
b_a	$1.5 \cdot 10^3$
k_a	$3 \cdot 10^{-3}$
R_{in}	0.1
R_f	$8 \cdot 10^4$

Table 8: Set Values

Point 2:

As stated before, the initial conditions chosen for this part were 10^{-8} for all the state variables. Other initial conditions could lead to better results, however, the improvement in solution would be more costly in aspects of clarity and efficiency of the code. Once the initial parameters and values were chosen, the system of equation already presented was introduced in a MATLAB integrator, for the indicated time interval $[0 \ 4.51]$ h.

The solutions obtained are given in Fig 14 for accelerometer, Fig. 15 and Fig. 16 for the solenoid.

	Absolute Value
$ \lambda_1 $	$4.68 \cdot 10^3$
$ \lambda_2 $	$10 \cdot 10^3$
$ \lambda_3 $	$4.38 \cdot 10^{-12}$
$ \lambda_4 $	$5.12 \cdot 10^{-2}$
$ \lambda_5 $	$8.98 \cdot 10^{-1}$

Table 9: System's Eigenvalues for last time step in absolute form

Fig. 17 depicts the evolution of Drag and Thrust in the time interval and it can be observed that the two have very similar behavior and appropriate values once steady state is reached. Firstly, the small values presented by the system are expected as the values of force acting on it, $(T - D)$ is very small, with an order of magnitude of 10^{-5} . The sinusoidal behavior is expected since drag, the driving parameter of the system, is defined by *sin* and *cosin* functions, Eq. 24. If a different value

for T were chosen, such that it would have been much larger than the drag, then the behavior of the system would partly lose its oscillatory form, as it was observed for different values. This is normal, as the system depends on the $T - D$ evolution.

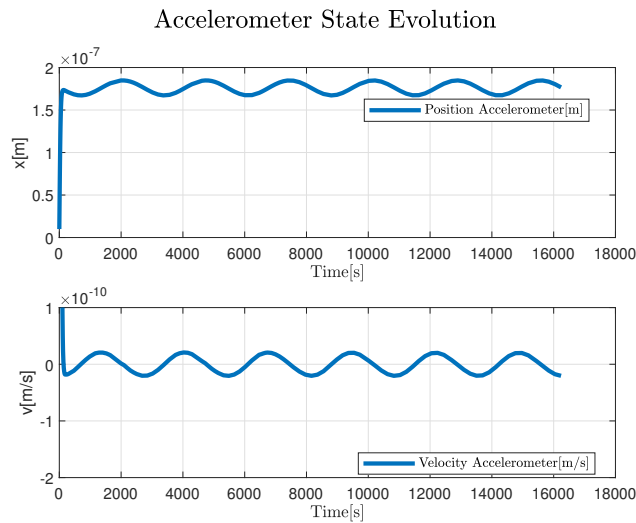
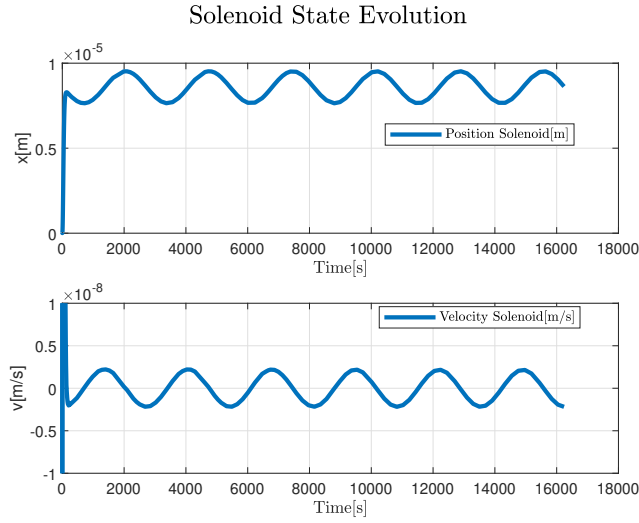
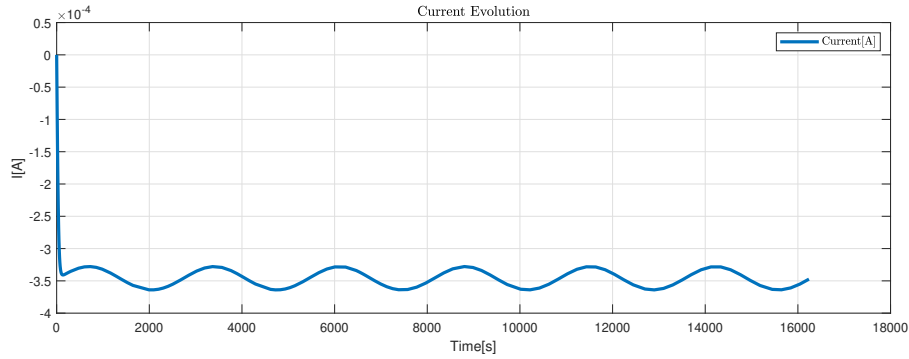
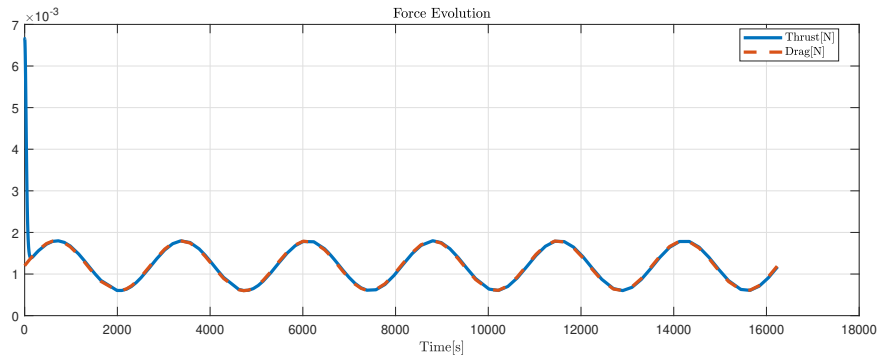


Figure 14: Accelerometer State Variables Evolution

Point 3:

**Figure 15:** Solenoid State Variables Evolution**Figure 16:** Solenoid Current Evolution**Figure 17:** Thrust and Drag Time Evolution

The ODE integrator used for this exercise was MATLAB's *ode15s*. Similarly to the previous exercise, the integrator was chosen to efficiently calculate the solutions. That is, the system is stiff, which was observed once its eigenvalues were calculated, with λ_1 and λ_2 being 3 orders of magnitude larger than the others, as presented in Table 9. The dominance of λ_1 and λ_2 indicates rapid transients in the solenoid dynamics, requiring a stiff solver to capture these accurately. Another reason would be the difference in time scales between the mechanical and electrical subsystems, leading to stiffness, which was observed also from the time required by different integrators to solve the simulation. As *ode15s* is a stiff integrator from MATLAB's library, it

manages the integration process of the system more efficiently and accurately, compared to the other non-stiff integrators, as already seen in Table 8. The tolerances were set to 10^{-9} to factor in the force introduced in the system once multiplied by m_a/M_{SC} , when acting on the seismic mass.

Part 2: Acausal Modeling

For the acausal Modeling a physical model of the ACS was implemented in Simscape/Simulink. The initial conditions were considered to be the same as in the causal model, implemented in MATLAB. The model was divided into 3 parts similar to the three components of the ACS. The Acausal Modeling part will be presented through the overall scheme and afterwards each component of the ACS.

1. Accelerometer
2. Solenoid
3. Ion Thruster

The entire model of the ACS system is shown in Fig. 19. As it can be observed, besides the 3 components, there is block created in Simulink, that represents the drag. This was created in Simulink because it is a function of time, and defined through mathematical expressions, thus it was more advantageous as Simscape does not include already defined trigonometry blocks.

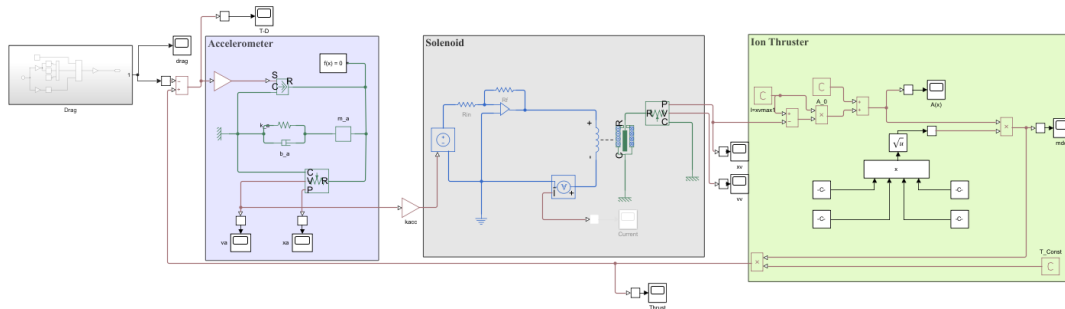


Figure 19: Acausal Model in Simscape

Thus, the drag was created in simulink and afterwards through the mean of *Simulink-PS converter*, the output was converted in Simscape.

Once the drag is calculated, it is subtracted from Thrust, and the input is introduced into the accelerometer. The Accelerometer, Fig. 18, is modeled as a spring-damper mass system. From this model, through a *Translation Motion Sensor*, the velocity and position of the mass are extracted.

The behavior of mass is afterwards introduced into the solenoid through a gain, that applies the accelerometer proportional coefficient to the velocity v_a . This behavior is then introduced into a controlled voltage source that powers the solenoid.

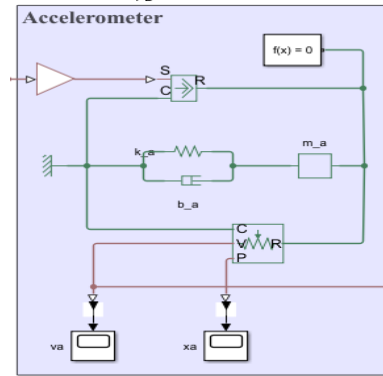


Figure 18: Accelerometer Simscape Model

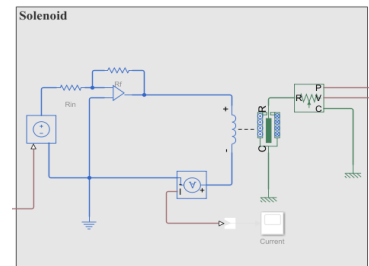


Figure 20: Model of the Amplifier and Solenoid.

The obtained voltage is then modulated and it powers the solenoid that was initialized using the values in Table 7 and Table 6.

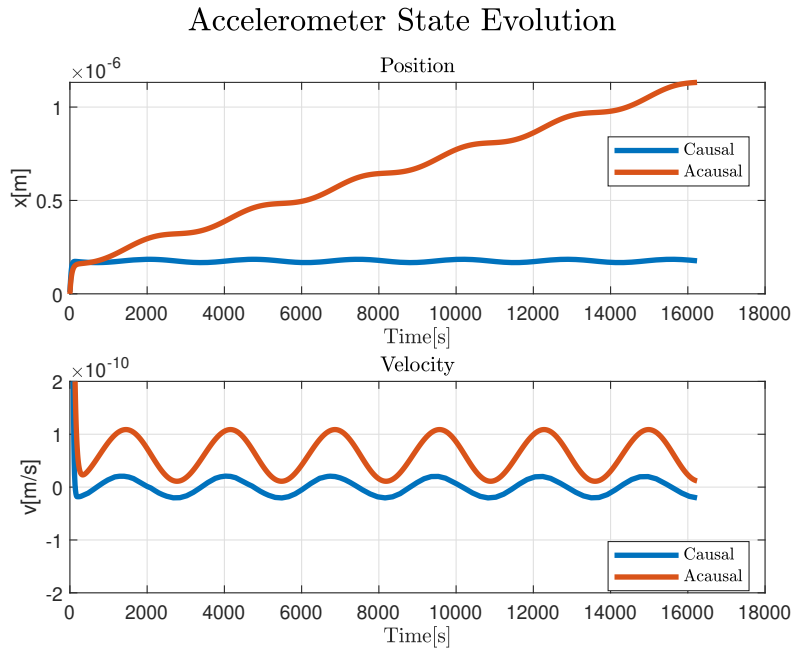


Figure 21: Difference between the Accelerometer states in Causal and Acausal Model

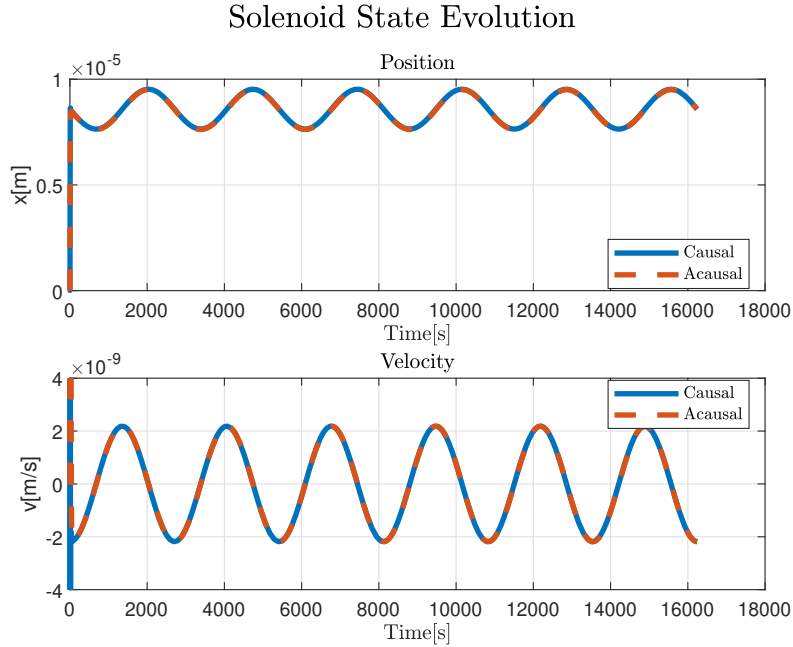


Figure 22: Difference between the Solenoid states in Causal and Acausal Model

The solenoid is modeled directly through the Simscape integrated block. From the solenoid, again through a *Translation Motion Sensor* the values of x_v and v_v are obtained. The evolution of the current is calculated directly from the electrical circuit using a *PS-Simulink converter* and a Scope. From the values obtained for the x_v , the equations of Area variation of the ion thruster, $(A(x_v))$. Afterwards, the remaining part of the Ion Thruster was modeled using causal blocks. An equivalent method was tested by using the Simscape acausal blocks, however, due

to the lack of data regarding the fluid dynamics from the reservoir and through the pipes such as discharge coefficient, specific enthalpy and so on, the results were slightly different than the desired ones. Thus, the Ion Thruster block was created using acausal models.

The integration is performed using *ode15s* integrator and tolerances of 10^{-7} . The results are presented in Fig. 21, Fig. 22 and Fig. 23. An important difference is observed in the acausal

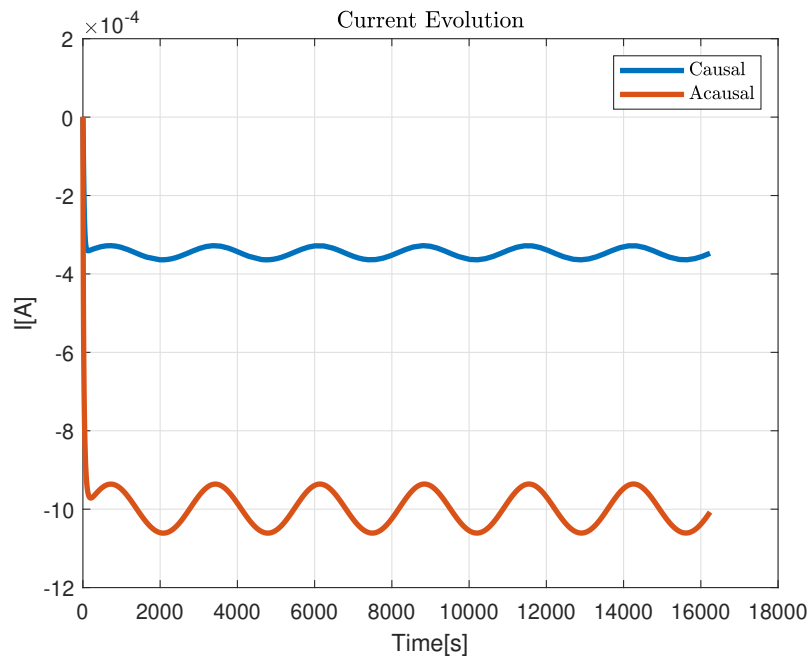


Figure 23: Difference between the Solenoid Current in Causal and Acausal Model

model with respect to the causal one. This is very noticeable especially in the accelerometer states evolution. The acausal model leads to a larger error, than the one implemented in Matlab. This can be based on the acausal blocks implemented in Simscape. Firstly, the solenoid block seems to contribute to the issue. Replacing it with a causal approach appears to resolve the problem. Specifically, the solenoid parameters used were the provided values, while any missing parameters were initialized with default Simscape values. This mismatch introduces a disturbance in the simulation, which is clearly visible in the evolution of x_a . Secondly, this leads to change in v_a that changes also the mean value, different from zero, which would lead to drift in x_a and a change in I . Moreover, if the electrical system were not dependent on velocity, the error propagation would be damped, and the results of the causal and acausal would be more similar.

References

- [1] R. Mukhiya, M. Garg, P. Gaikwad, et al. *Electrical equivalent modeling of MEMS differential capacitive accelerometer*. In: Microelectronics Journal. Vol. 99, pp. 104770, 2020. <https://doi.org/10.1016/j.mejo.2020.104770>